

# Beyond Clean Text: Evaluating Encoder and Decoder Robustness for Bangla Event Detection in Noisy Text

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## Abstract

Event detection (ED) systems are typically evaluated on clean, curated text, leaving their robustness to real-world noise largely unexplored, particularly for low-resource languages such as Bangla. We introduce a generalized Bangla news event ontology and a benchmark comprising 9,979 annotated sentences across 40 event subtypes, spanning clean news text, real-world Automatic Speech Recognition (ASR) transcripts, and orthographically corrupted text. We systematically evaluate fine-tuned encoder-only models (BanglaBERT and XLM-R) alongside instruction-tuned decoder-only large language models (Llama 3 and Gemma 3). Our results reveal a clear architectural trade-off: encoder models achieve higher performance on clean text but degrade substantially under noise, whereas decoder-only LLMs are markedly more robust, particularly when event triggers are corrupted. We further show that embedding annotation guidelines during instruction tuning establishes a higher performance baseline on noisy text but yields inconsistent reductions in performance degradation across noisy conditions. Finally, model scaling consistently improves the robustness of decoder-only LLMs, while combined training on clean and noisy data serves as an effective regularization strategy that disproportionately benefits encoder architectures, significantly narrowing the robustness gap.

## 1 Introduction

The task of event detection (ED) identifies and categorizes events in natural language (Ahn, 2006). It is a fundamental component of information extraction, enabling downstream applications such as emergency monitoring, where timely and accurate extraction of structured events is critical. Despite its importance, most ED research has focused on clean, carefully curated text (Wang et al., 2020; Pouran Ben Veyseh et al., 2022; Yao et al.,

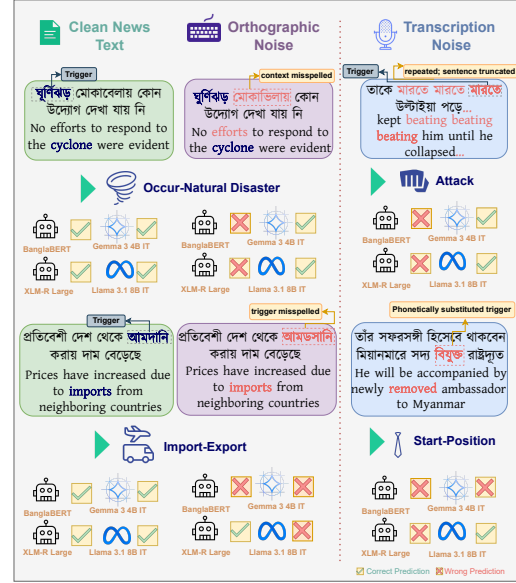


Figure 1: Real-world noise substantially degrades event detection, even when the event trigger itself remains unchanged. Clean examples are drawn from our Clean Test set, while their counterparts are generated using simulated orthographic noise. ASR examples are collected from real Bangla news video transcripts and independently annotated using the same event ontology, introducing naturally occurring transcription artifacts and out-of-distribution vocabulary.

2022). In practice, however, textual data is inherently noisy, and even minor typographical errors or ASR transcription mistakes can cause event detection systems to miss critical events (Figure 1).

As a low-resource language, Bangla remains underexplored for event detection. Existing studies primarily focus on clean text and narrow domains such as violent incidents (Khandokar et al., 2020; Dey et al., 2021; Ali Khandokar et al., 2025), disasters (Dave et al., 2021), or crime-related events (Hossain et al., 2025). Moreover, there is no generalized Bangla news event ontology comparable to ACE 2005 (Walker, Christopher et al., 2006) for

trigger-based event detection, thereby limiting the systematic study of the task. Research on noisy Bangla text is similarly limited and has largely been restricted to sentence-level sentiment analysis (Islam et al., 2021; Elahi et al., 2024), whereas event detection requires fine-grained token-level identification and classification of event triggers.

Historically, event detection has been dominated by BERT-based encoder models (Wang et al., 2020; Pouran Ben Veyseh et al., 2022; Huang et al., 2024). In contrast, decoder-only LLMs have only recently been adapted to structured extraction through techniques such as code-based representations (Wang et al., 2023), incorporation of sentence-level contextual information (Al Monsur et al., 2026) and instruction tuning with annotation guidelines (Srivastava et al., 2025). Given these advances, an important question emerges regarding how these models compare with their BERT-based counterparts under real-world noisy conditions, particularly for a low-resource language such as Bangla.

To answer this question, we annotate 5,320 sentences from Bangla news articles and 4,659 ASR transcripts from Bangla news videos using same event ontology. We evaluate both encoder-only and decoder-only models for event detection, formulating the latter as a structured code generation task (Wang et al., 2023) with annotation guidelines optionally embedded in the prompt (Sainz et al., 2024; Srivastava et al., 2025). In addition, we simulate orthographic noise using the error generation algorithm of Sifat et al. (2020), which was developed by analyzing common Bangla writing patterns and typographical errors, and evaluate model robustness under varying noise levels.

In summary, our contributions are as follows:

- We develop a generalized Bangla news event ontology and release a human-annotated dataset comprising 9,979 sentences from clean news articles and noisy ASR transcripts, containing 8,041 event mentions across 40 event subtypes.
- We show that although encoder-based models achieve higher performance on clean text, decoder-only LLMs are substantially more robust to noise, particularly when event triggers are corrupted.
- We find that instruction tuning with annotation guidelines consistently improves over-

all event detection performance but yields limited robustness gains, whereas combined training on clean and ASR data substantially improves encoder robustness and narrows the gap with decoder-only LLMs.

Code and dataset to reproduce our experiments are available.<sup>1</sup>

## 2 Related Work

Early event detection systems relied on feature engineering and statistical learning techniques (Ahn, 2006; Patwardhan and Riloff, 2009; Hong et al., 2011; Li et al., 2013). More recently, pretrained Transformer encoders such as BERT have become the dominant paradigm, substantially advancing trigger detection performance (Nguyen et al., 2021; Wang et al., 2020; Pouran Ben Veyseh et al., 2022; Huang et al., 2024). Alongside these architectural advances, recent work has also focused on developing event detection datasets spanning diverse domains (Kim et al., 2009; Sims et al., 2019; Le and Nguyen, 2021; Yao et al., 2022) and languages (Pouran Ben Veyseh et al., 2022; Touileb et al., 2024). Large-scale resources such as MAVEN (Wang et al., 2020), RAMS (Ebner et al., 2020), and the TextEE benchmark (Huang et al., 2024) have become standard evaluation platforms for modern event detection research.

Compared with the extensive literature on English event detection, Bangla has received considerably less attention. Existing studies have primarily developed task-specific datasets and models targeting individual application domains, including violent incidents (Khandokar et al., 2020; Dey et al., 2021; Ali Khandokar et al., 2025), disasters (Dave et al., 2021), and crime-related news (Hossain et al., 2025). Moreover, publicly available resources do not provide a generalized news-domain ontology comparable to ACE 2005 (Walker, Christopher et al., 2006), making standardized evaluation across diverse event categories difficult. Studies on noisy Bangla text are likewise limited and have focused primarily on sentence-level tasks such as sentiment analysis (Islam et al., 2021; Elahi et al., 2024) rather than token-level trigger identification.

The rapid progress of decoder-only LLMs has recently motivated their application to event detection. Recent work reformulates the task as

<sup>1</sup><https://github.com/sijantanvir/Bangla-Noisy-Event-Eval>

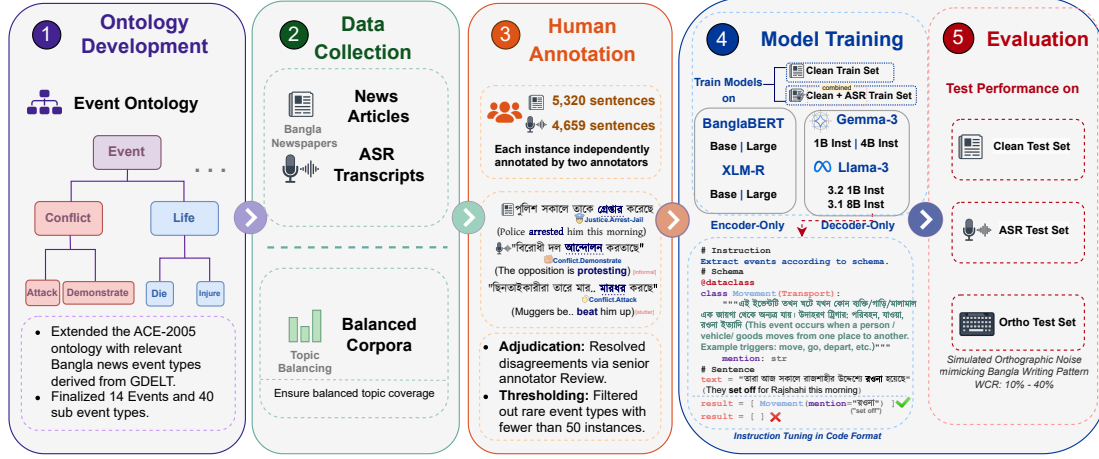


Figure 2: Overview of the dataset construction, training, and evaluation pipeline. We develop a generalized Bangla news event ontology, collect topic-balanced news and ASR corpora, and annotate both using the same event schema. Encoder-only and decoder-only models are trained under two settings: Clean and Combined (Clean + ASR) Training Set. Decoder-only LLMs are instruction-tuned using code-format prompts with and without annotation guidelines embedded as Python docstrings. Robustness is evaluated on the Clean, real-world ASR transcription noise, and simulated orthographic noise test sets.

structured code generation (Wang et al., 2023), explores context-aware encoding (Al Monsur et al., 2026), and employs instruction tuning with annotation guidelines to improve schema understanding and cross-schema generalization (Srivastava et al., 2025). Generative formulations have also been applied to complex event argument extraction (Sharif et al., 2024).

Beyond event extraction, annotation guidelines have also been shown to improve other information extraction tasks, including Named Entity Recognition (Sainz et al., 2024) and Relation Extraction (Pang et al., 2023). However, these methods have almost exclusively been evaluated on clean English benchmarks, leaving the robustness of encoder-only and decoder-only models, as well as the impact of annotation guidelines during instruction tuning, under realistic noisy conditions largely unexplored.

In this work, we address these gaps using Bangla as a representative low-resource language. We introduce a generalized Bangla news event ontology, construct a benchmark comprising clean news articles, real-world ASR transcripts, and simulated orthographic noise, and systematically compare fine-tuned encoder-only and instruction-tuned decoder-only models. We further investigate whether annotation guidelines improve model performance and robustness under noisy conditions.

## 3 Benchmark Development

### 3.1 Task Formulation

We evaluate event detection across two distinct modeling paradigms: Token Classification, for encoder-only architectures and Structured Sequence Generation, for decoder-only large language models.

Formally, let an input text sequence of length  $n$  be defined as  $X = [x_1, x_2, \dots, x_n]$ . Our ontology consists of a predefined set of event types  $\mathcal{E}$ . The goal of the event detection system is to extract a set of event instances  $Y$  from  $X$ . Each extracted event instance is defined as a tuple  $(i, j, t)$ , where  $[x_i, \dots, x_j]$  represents the continuous text span that triggers the event, and  $t \in \mathcal{E}$  is the predicted event type. To systematically evaluate the capabilities of different architectures, the extraction of  $Y$  is formulated through two paradigms:

**Event Detection as Token Classification.** The task is framed as a token-level classification problem. A model processes the input text  $X$  and assigns a label  $y_k$  to each token. The model must simultaneously perform *Trigger Identification* (locating the boundaries  $i$  and  $j$  of the trigger span) and *Trigger Classification* (mapping the identified span to an event type  $t \in \mathcal{E}$ ).

**Event Detection as Structured Sequence Generation.** For generative LLMs, the task is reframed as conditional sequence generation. Given the in-

put sentence  $X$ , a natural language task instruction  $I$ , and the target event schema  $E_t$  for a specific event type  $t \in \mathcal{E}$ , we construct a unified prompt  $P$ :

$$P = [I \oplus E_t \oplus X] \quad (1)$$

where  $\oplus$  denotes string concatenation. The instruction  $I$  dictates the extraction rules and desired structured output format, while  $E_t$  provides the semantic definition of the target event. The model is trained to autoregressively generate the structured representation of  $Y$ , explicitly handling both the extraction of valid triggers and the rejection of schemas not present in  $X$ .

### 3.2 Ontology Development

We adopted the ACE 2005 (Walker, Christopher et al., 2006) event ontology as the foundation of our schema and extended it with event types relevant to Bangladeshi news but not explicitly covered by ACE (e.g., disasters, festivals, socio-economic events). To identify these event types, we analyzed Bangladesh-specific CAMEO event codes and GKG themes from GDELT (Leetaru and Schrodt, 2013) between 2023 and 2025.

The resulting ontology initially comprised 16 top-level event categories and 70 sub-types. After annotation, event types with fewer than 50 instances were removed to improve evaluation reliability, yielding a final ontology of 14 main event categories and 40 sub-types (Appendix A). Definitions and representative examples for all event types and subtypes are provided in Appendix C.

#### 3.2.1 Data Collection

**News Text** To construct the clean text corpus, we utilized the Bangla News Article Dataset (BNAD) (Saad et al., 2024), which contains category-wise collections of news articles sourced from Bangla Online Newspapers. We collected articles from three major Bangladeshi newspapers: Dhaka Tribune Bangla, The Daily Ittefaq, and Daily Janakantha. Articles were sampled from categories aligned with the event categories in our ontology to ensure balanced coverage across diverse domains. Coverage of topics in the dataset are described in Appendix A.

**ASR Transcript** To construct our real-world noisy ASR corpus, we curated publicly available news broadcasts from the official YouTube channels of three of the most popular Bangladeshi television networks: Jamuna TV, Somoy TV, and In-

dependent Television. We extracted the automatically generated Bangla speech transcripts associated with these broadcasts. Utilizing these ASR transcripts provides a naturally occurring source of noisy Bangla text. This data inherently captures colloquial vocabulary, dialectal verb inflections, homophone substitutions, and transcription errors, accurately reflecting the morphological and lexical challenges encountered by event extraction systems in real-world spoken media.

### 3.3 Dataset Quality Control

**Annotation Procedure** We developed detailed annotation guidelines based on the proposed ontology, including event definitions, representative trigger examples, boundary annotation instructions, and common edge cases. Six annotators were selected through a screening test based on these guidelines. Annotation was conducted using the academic version of Label Studio (HumanSignal, 2020).

**Inter-Annotator Agreement** Each sentence was independently annotated by two annotators. Inter-annotator agreement, measured using Cohen’s  $\kappa$  (Cohen, 1960), was 0.72.

**Adjudication and Quality Control** Disagreements were resolved through peer discussion followed by senior annotator adjudication. Sentences with unresolved ambiguity were excluded, resulting in the removal of approximately 18% of annotated sentences.

### 3.4 Dataset Statistics

In total, the finalized benchmark corpus comprises 9,979 annotated sentences, including 5,320 sentences from news articles and 4,659 sentences from ASR transcripts, containing a total of 8,041 event mentions. Dataset statistics are summarized in Table 1.

| Source         | # Annotated Sentences | # Event Mentions | Avg. Words /Instance |
|----------------|-----------------------|------------------|----------------------|
| News Article   | 5320                  | 4406             | 14.1                 |
| ASR Transcript | 4659                  | 3635             | 11.5                 |
| <b>Total</b>   | <b>9979</b>           | <b>8041</b>      | <b>12.9</b>          |

Table 1: Dataset Statistics

### 3.5 Event Class Distribution

Figure 3 illustrates the distribution of event instances across the 40 event subtypes in the News

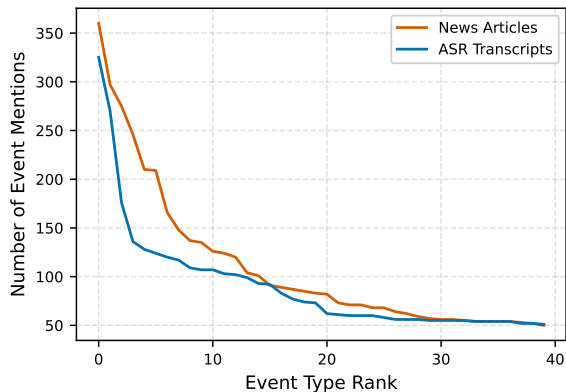


Figure 3: Frequency distribution of the 40 event subtypes in the News and ASR corpora, showing the long-tail nature of both datasets. Rankings are computed independently for each corpus.

Text and ASR Transcript corpora. Both subsets exhibit similar long-tail distributions, reflecting our topic-balancing effort. Detailed class distributions are provided in Figure 8 of Appendix A.

### 3.6 Orthographic Noise Test Data

Although our ASR corpus captures errors arising from acoustic ambiguity, pronunciation variation, and missing or incorrectly recognized tokens, it does not reflect the lexical and character-level spelling errors commonly found in user-generated Bangla text. These errors are primarily caused by phonetic substitutions, keystroke errors, spelling mistakes, and incorrect handling of *Juktakkhar* (consonant conjuncts). Consequently, robustness to ASR noise does not necessarily imply robustness to orthographic noise.

Rather than collecting and re-annotating a large corpus of naturally occurring spelling errors, we simulate orthographic noise using the algorithm of Sifat et al. (2020). The algorithm was developed through an analysis of real Bengali typing patterns, including phonetic substitutions, frequently mispressed key clusters, insertion errors, and *Juktakkhar*-specific rules. As a result, it produces realistic Bangla spelling errors while preserving the original event annotations.

To evaluate robustness under increasing corruption, we apply the algorithm to the Clean Test set at Word Corruption Rates (WCR), defined as  $x \in \{10, 20, 30, 40\}$ . At each level, approximately  $x\%$  of the words are randomly selected and transformed using the error generation algorithm. Corruption is applied cumulatively, such that each higher WCR retains all previously cor-

rupted words while introducing errors into additional randomly selected words.

## 4 Experiments

### 4.1 Experimental Setup

**Datasets and Evaluation Splits** We evaluate models on three test sets: **Clean Test**, **ASR Test**, and **Ortho Test**, the latter generated from the Clean Test set by injecting orthographic noise at varying Word Corruption Rates (WCR). Since sentences may contain multiple event triggers following a long-tail distribution, we partition the data using iterative multi-label stratification (Sechidis et al., 2011; Szymaski and Kajdanowicz, 2017) to preserve class distributions across splits. The Clean corpus is divided into train/validation/test (70/15/15), while the ASR corpus is divided into train/validation/test (70/10/20). Table 2 summarizes the resulting dataset sizes.

| Dataset                | Train & Val | Test             |
|------------------------|-------------|------------------|
| News Text              | 4504        | 816              |
| ASR Transcripts        | 3705        | 954              |
| Combined (Clean + ASR) | 8209        | –                |
| Ortho Test             | –           | 816 <sup>†</sup> |

Table 2: Raw sentence counts across dataset splits.

<sup>†</sup>Indicates the count per discrete WCR level

**Models** For the encoder-only baselines, we employ BanglaBERT (Base: 110M, Large: 335M) (Bhattacharjee et al., 2022) and XLM-RoBERTa (Base: 270M, Large: 550M) (Conneau et al., 2020). BanglaBERT represents the state-of-the-art monolingual pretrained model for Bangla NLP, while XLM-R provides a strong multilingual baseline.

For the generative LLM baselines, we evaluate several open-weight LLMs across different parameter scales and multilingual support. Specifically, we consider models from the Llama family, including Llama 3.1 8B IT and Llama 3.2 1B IT (Dubey et al., 2024), together with Google’s Gemma 3 4B IT and Gemma 3 1B IT (Gemma Team et al., 2025).

**Model Training** Encoder-only models are trained for token classification, whereas decoder-only LLMs are instruction-tuned for event detection. We employ targeted negative sampling (NS), where each positive event query is

augmented with five randomly selected negative event types (1:5 ratio), providing contrastive supervision for distinguishing event types and suppressing hallucinated predictions (Srivastava et al., 2025). To evaluate the effect of annotation guidelines, we consider two prompting settings: **w/ Guide**, where event definitions are embedded as Python docstrings, and **w/o Guide**, where only the event type name is provided. Detailed prompt templates and training hyperparameters are provided in Appendix B.1 and B.2.

**Evaluation Protocol** Following Srivastava et al. (2025), we adopt a one-vs-all inference strategy for decoder-only LLMs, pairing each test sentence with every event type independently and aggregating predictions across all 40 event types. This formulation provides explicit supervision for absent event types, reducing hallucinated event predictions while enabling a fair comparison with encoder-only token classification models.

**Evaluation Metric** We report Macro-F1 as the primary metric because it equally weights all event types regardless of class frequency, making it more appropriate for our long-tailed event distribution than Micro-F1 (Al Monsur et al., 2026).

## 4.2 Evaluation of Language Models

**LLMs exhibit greater robustness to noise than encoder-based models.** To quantify robustness, we measure the performance degradation due to noise as  $\Delta F1 = F1_{\text{Clean}} - F1_{\text{Noisy}}$ , where  $F1_{\text{Noisy}} = \frac{1}{2}(F1_{\text{ASR}} + F1_{\text{Ortho-Avg}})$  averages performance on real-world ASR transcripts and simulated orthographic corruption.

Figure 4 reveals a clear distinction between encoder-based models and generative LLMs. Although encoder models achieve higher clean-text performance, they are substantially more sensitive to noise. For example, BanglaBERT Large attains the highest clean Macro F1 (68.9) but drops by 14.6 points under noisy conditions, while XLM-R Large degrades by 11.4 points. In contrast, LLMs remain considerably more robust. Llama 3.1 8B IT (w/ Guide) achieves a clean Macro F1 of 57.6 with only a 2.7 point drop, and Gemma 3 4B IT (w/ Guide) degrades by just 4.0 points.

**Instruction tuning w/ Guide improves performance but yields inconsistent reductions in clean-to-noisy performance drop** As shown in Figure 5 and Table 3, instruction tuning w/ Guide

consistently improves Macro F1 on both the Clean and ASR test sets across all evaluated LLM families, despite training exclusively on clean news text. For example, Gemma 3 4B IT improves from 55.2 to 60.6 Macro F1 on the Clean Test set, with similar gains across other models. However, its effect on robustness is inconsistent. For Gemma 3 4B IT and Llama 3.2 1B IT, the average performance drop ( $\Delta F1$ ) increases from 2.9 to 4.0 and from 4.9 to 6.4 points, respectively. In contrast, Llama 3.1 8B IT benefits substantially, reducing its degradation from 6.9 to 2.7 points.

**Noise combined training enhances encoder robustness** Training on a mixture of clean and noisy ASR data improves robustness for both encoder-based models and generative LLMs, although the gains differ across architectures. Encoder models benefit most: BanglaBERT Large reduces its average performance drop ( $\Delta F1$ ) from 14.6 to 7.9 points, while XLM-R Large improves from 11.4 to 5.3 points, with little change in clean-test performance. In contrast, generative LLMs improve both clean performance and robustness. For example, Gemma 3 4B IT (w/ Guide) reduces its average performance degradation from 4.0 to 3.3 points, while Llama 3.1 8B IT (w/ Guide) reduces its degradation from 2.7 to 1.0 points, alongside an improvement in its clean-test score. Overall, combined training is particularly beneficial for encoder architectures, substantially narrowing the robustness gap with generative LLMs.

**Model scaling benefits decoder-only LLMs more than encoders** Parameter scaling reveals a clear architectural divergence in robustness. For decoder-only models, larger models consistently exhibit smaller noise-induced performance drops. For example, scaling from Llama 3.2 1B IT to Llama 3.1 8B IT reduces the average Macro-F1 degradation from 6.4 to 2.7 points under clean training (w/ Guide). A similar trend is observed for the Gemma family, where the 4B model consistently outperforms its 1B counterpart in robustness. In contrast, scaling encoder models from Base to Large improves clean-text performance but does not consistently improve robustness. Under clean training, BanglaBERT-Large and XLM-R-Large incur larger average performance drops than their Base counterparts (14.6 vs. 13.3 and 11.4 vs. 10.4 points, respectively).

| Architecture | Model            | Training Condition | Macro F1   |          |                   |                |                  |
|--------------|------------------|--------------------|------------|----------|-------------------|----------------|------------------|
|              |                  |                    | Clean Test | ASR Test | Ortho Test (Avg.) | $\Delta_{ASR}$ | $\Delta_{Ortho}$ |
| Decoder-Only | Gemma 3 4B IT    | Clean w/ Guide     | 60.57      | 59.53    | 53.66             | 1.04           | 6.91             |
|              |                  | Clean w/o Guide    | 55.19      | 53.94    | 50.58             | 1.25           | 4.61             |
|              |                  | Combined w/ Guide  | 65.60      | 65.43    | 59.14             | 0.17           | 6.46             |
|              | Gemma 3 1B IT    | Clean w/ Guide     | 52.27      | 48.81    | 45.11             | 3.46           | 7.16             |
|              |                  | Clean w/o Guide    | 48.88      | 45.75    | 44.31             | 3.13           | 4.57             |
|              |                  | Combined w/ Guide  | 59.07      | 54.58    | 50.40             | 4.49           | 8.67             |
|              | Llama 3.1 8B IT  | Clean w/ Guide     | 57.58      | 55.23    | 54.57             | 2.35           | 3.01             |
|              |                  | Clean w/o Guide    | 55.27      | 46.92    | 49.86             | 8.35           | 5.41             |
|              |                  | Combined w/ Guide  | 64.95      | 66.39    | 61.60             | -1.44          | 3.35             |
|              | Llama 3.2 1B IT  | Clean w/ Guide     | 54.64      | 48.07    | 48.38             | 6.57           | 6.26             |
|              |                  | Clean w/o Guide    | 47.03      | 43.89    | 40.42             | 3.14           | 6.61             |
|              |                  | Combined w/ Guide  | 56.37      | 59.33    | 51.93             | -2.96          | 4.44             |
| Encoder-Only | BanglaBERT Base  | Clean              | 67.35      | 56.79    | 51.35             | 10.56          | 16.00            |
|              |                  | Combined           | 68.49      | 68.58    | 52.02             | -0.09          | 16.47            |
|              | BanglaBERT Large | Clean              | 68.91      | 58.34    | 50.25             | 10.57          | 18.66            |
|              |                  | Combined           | 68.84      | 69.38    | 52.41             | -0.54          | 16.43            |
|              | XLM-R Base       | Clean              | 67.35      | 55.46    | 58.54             | 11.89          | 8.81             |
|              |                  | Combined           | 68.29      | 66.20    | 58.80             | 2.09           | 9.49             |
|              | XLM-R Large      | Clean              | 70.61      | 56.92    | 61.43             | 13.69          | 9.18             |
|              |                  | Combined           | 71.16      | 69.34    | 62.33             | 1.82           | 8.83             |

Table 3: Macro-F1 performance of encoder and decoder architectures across clean and noisy test sets.  $\Delta_{ASR}$  and  $\Delta_{Ortho}$  denote the absolute performance degradation relative to the clean test set (Clean – Noisy). Smaller  $\Delta$  values indicate greater robustness, while negative values indicate improved performance on the noisy test set compared with the clean baseline.

**Progressive orthographic noise induces monotonic performance decline, modulated by training strategy.** As Word Corruption Rate (WCR) increases from 0% to 40%, both generative LLMs and encoder-based models exhibit a monotonic decline in Macro F1 (Figure 5). However, the rate of degradation varies across architectures and training strategies. Combined training consistently improves robustness by flattening the degradation curve, with the largest gains observed for encoder models. For generative LLMs, combined training also improves absolute performance across most corruption levels. Across all evaluated model families, w/ Guide variants consistently outperform their w/o Guide counterparts on both clean and corrupted inputs.

**LLMs excel under trigger corruption, while encoders remain competitive under context corruption.** To complement the aggregate analysis, we perform paired McNemar’s exact tests at WCR = 40%, the most challenging evaluation setting. Predictions are partitioned according to

whether the event trigger itself is corrupted (where surrounding context may also contain corruption) or whether only the surrounding context is corrupted while the trigger remains intact. Statistical significance is computed using strict trigger identification.

Figure 6 reveals a clear distinction between the two settings. Under trigger corruption, decoder-only LLMs, particularly larger models, significantly outperform encoder-based architectures, demonstrating greater robustness to corrupted trigger mentions. In contrast, under context corruption, the performance gap narrows substantially. Most comparisons are no longer statistically significant, and among the significant cases, XLM-R frequently outperforms the smaller LLMs. These results suggest that the primary advantage of decoder-only LLMs lies in recovering corrupted trigger mentions, whereas encoder models remain highly effective when only the surrounding context is noisy.

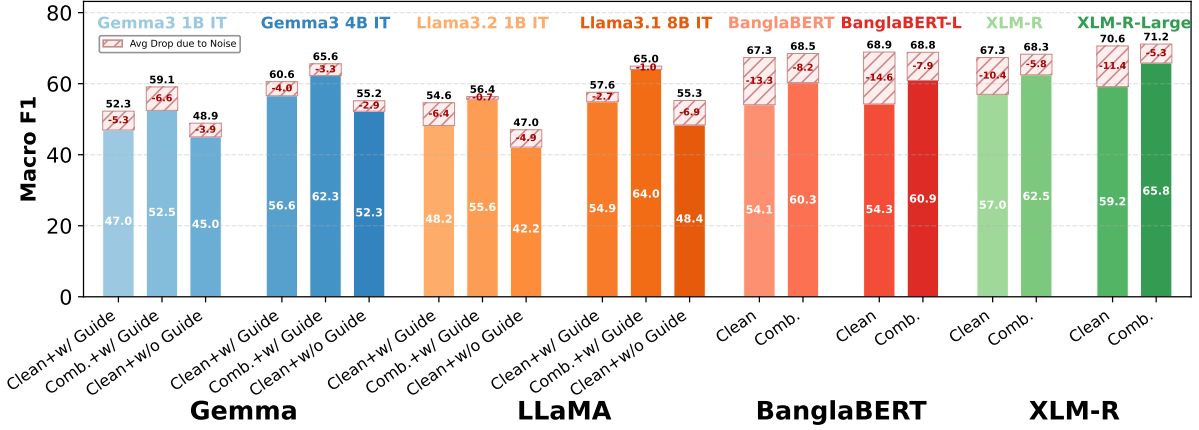


Figure 4: Macro-F1 decomposition across model architectures and training settings. Bar height denotes Clean Test Macro-F1; the solid region shows average performance retained across ASR and orthographic test sets, and the hatched region the corresponding performance drop.

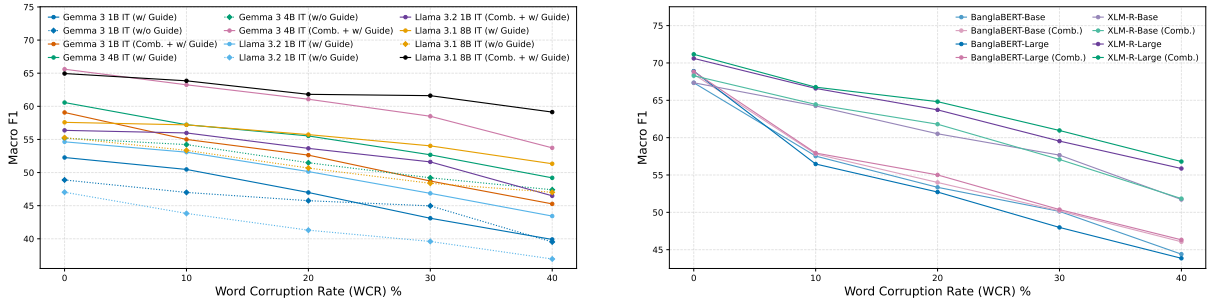


Figure 5: Comparison of performance degradation between Generative LLMs (left) and Encoder architectures (right) under varying levels of orthographic noise.

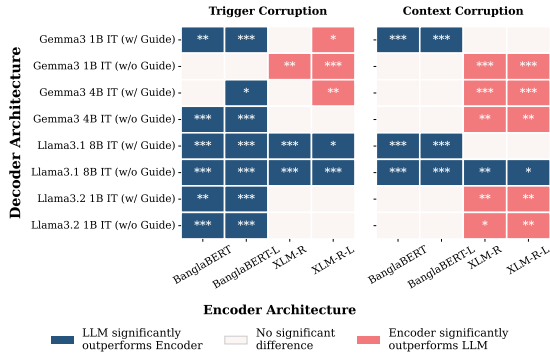


Figure 6: Pairwise McNemar's significance tests at WCR = 40%. Left: Trigger Corruption (with accompanying context corruption). Right: Context-Only corruption (Trigger intact). Significance levels: \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ .

## 5 Conclusion

We presented the first comprehensive study of language model robustness for Bangla event detection under real-world ASR transcription errors and simulated orthographic noise. We in-

troduced a generalized Bangla news event ontology and released a dataset of 9,979 human-annotated sentences spanning clean news, ASR transcripts, and synthetic orthographic corruption. Our experiments reveal a clear architectural trade-off: encoder-only models achieve stronger performance on clean text but degrade substantially under ASR and orthographic noise, whereas decoder-only LLMs are considerably more robust, particularly when event triggers are corrupted. We further show that instruction tuning with annotation guidelines consistently improves overall performance but provides limited gains in robustness, while combined training on clean and noisy text substantially improves encoder robustness. These findings highlight the importance of evaluating event detection beyond clean-text benchmarks and establish strong baselines for robust event detection in low-resource settings.

## Limitations

Although we carefully designed our experiments to ensure a fair comparison between encoder-based models and decoder-only LLMs, our study has several limitations.

**Strict Span-Based Evaluation.** Following standard event detection benchmarks, we evaluate using Strict Macro-F1, which requires an exact match between predicted and gold trigger spans and event types. This metric measures precise trigger identification, the primary objective of event detection, and enables direct comparison across encoder-only and decoder-only models. Because decoder-only LLMs generate trigger mentions autoregressively, they may occasionally produce semantically appropriate mentions with slightly different surface boundaries, which are counted as incorrect under the strict criterion. Accordingly, our results should be interpreted as reflecting exact trigger extraction rather than semantic equivalence of generated mentions.

**Domain and Task Scope.** This study is restricted to the news domain, using datasets constructed from formal news articles and their same-domain ASR transcripts. Consequently, the reported robustness characteristics may not generalize to general-domain event detection, which may involve other event types and exhibit substantially different linguistic properties and noise distributions. Furthermore, our investigation is limited to event detection, namely identifying event trigger mentions and classifying their event types, as annotating event arguments, participant roles, temporal expressions, and other event attributes would require substantial annotation effort and computational resources. Future research should investigate whether the robustness trends observed for event detection extend to argument extraction and complete event extraction pipelines.

## Ethical Considerations

Annotators were compensated for their work at a rate consistent with local standards for similar annotation tasks. Prior to participation, annotators were informed about the purpose of the study and how their contributions would be used in the dataset. All news articles and Automatic Speech Recognition (ASR) transcripts were collected from publicly available channels. The dataset was compiled strictly for non-commercial,

academic research purposes under standard fair use principles. Any named entities or individuals present in the text are public figures or subjects of public record, and no sensitive Personally Identifiable Information (PII) of private citizens was targeted or exposed.

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## A Data Collection and Ontology Development

### A.1 Topic Coverage

To ensure comparable topic-coverage across clean text and ASR transcripts, we curated both corpora from the same set of news domains. For the clean text corpus, we selected articles from the BNAD (Saad et al., 2024) dataset, specifically using content from three major Bangladeshi newspapers: *Dhaka Tribune Bangla*, *The Daily Ittefaq*, and *Daily Janakantha*. We focused on seven major news domains:

- রাজনীতি (Politics)
- অর্থনীতি (Economy)
- স্বাস্থ্য (Health)
- শিক্ষা (Education)
- প্রযুক্তি/টেক/বিজ্ঞান ও প্রযুক্তি (Technology/Science)
- বাণিজ্য (Commerce)
- চাকরি (Jobs)

For the ASR corpus, we curated publicly available Bangla news broadcasts from the official YouTube channels of *Jamuna TV*, *Somoy TV*, and *Independent Television*, and extracted their automatically generated Bangla speech transcripts. To mirror the clean text corpus, we collected broadcasts from category-specific playlists corresponding to the same seven news domains whenever available.

### A.2 Ontology Development

To construct a generalized Bangla news event ontology, we analyzed a large-scale corpus of Bangladeshi news using the Global Database of Events, Language, and Tone (GDELT) through Google BigQuery and queried the Global Knowledge Graph (GKG) to extract historical event records related to Bangladesh from 2023 to 2025 (Figure 7). Based on this corpus analysis, we identified event phenomena that are prevalent in Bangladeshi news and introduced locally relevant event types and subtypes to complement the widely adopted ACE 2005 ontology. This process ensured compatibility with established event types while capturing events that are underrepresented or absent in ACE 2005 but frequently occur in Bangladeshi news. The resulting ontology comprises 40 event subtypes organized into generalized event categories. The complete list of event

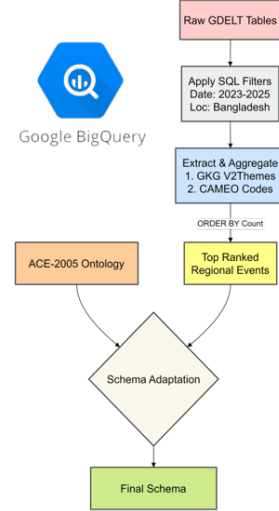


Figure 7: Overview of the ontology construction process. We analyze historical Bangladeshi news events from GDELT using Google BigQuery and the Global Knowledge Graph (GKG), adapt relevant event types, and integrate them with ACE 2005 to construct a generalized Bangla news event ontology.

types and subtypes, together with their sources, is presented in Table 4. The distribution of annotated event instances across the News and ASR corpora is shown in Figure 8.

Table 4: Event types and sub types in the dataset

| Event Type    | Sub Type                    | Source   |
|---------------|-----------------------------|----------|
| Contact       | Meet                        | ACE 2005 |
|               | Phone-Write                 | ACE 2005 |
| Conflict      | Attack                      | ACE 2005 |
|               | Demonstrate                 | ACE 2005 |
| Crime         | Commit-Blue-Collar-Crime    | GDELT    |
|               | Commit-White-Collar-Crime   | GDELT    |
| Disaster      | Occur-Man-Made-Disaster     | GDELT    |
|               | Occur-Natural-Disaster      | GDELT    |
| Festival      | Celebrate-Cultural-Festival | GDELT    |
|               | Observe-Religious-Festival  | GDELT    |
| Governance    | Approve                     | GDELT    |
|               | Ban                         | GDELT    |
|               | Decide                      | GDELT    |
|               | Gazette                     | GDELT    |
|               | Investigate                 | GDELT    |
|               | Reform                      | GDELT    |
|               | Support                     | GDELT    |
| Health        | Outbreak                    | GDELT    |
|               | Serve-Patients              | GDELT    |
| Justice       | Arrest-Jail                 | ACE 2005 |
|               | Charge-Indict               | ACE 2005 |
|               | Deliver-Verdict             | ACE 2005 |
|               | Sue                         | ACE 2005 |
|               | Trial-Hearing               | ACE 2005 |
| Life          | Die                         | ACE 2005 |
|               | Injure                      | ACE 2005 |
| Movement      | Transport                   | ACE 2005 |
| Personnel     | Elect                       | ACE 2005 |
|               | End-Position                | ACE 2005 |
|               | Start-Position              | ACE 2005 |
| Socio-Economy | Disrupt                     | GDELT    |
|               | Graduate                    | GDELT    |
|               | Grow                        | GDELT    |
|               | Import-Export               | GDELT    |
|               | Invest                      | GDELT    |
|               | Recruit                     | GDELT    |
|               | Trade                       | GDELT    |
| Technology    | Launch-Service              | GDELT    |
| Transaction   | Transfer-Money              | ACE 2005 |
|               | Transfer-Ownership          | ACE 2005 |

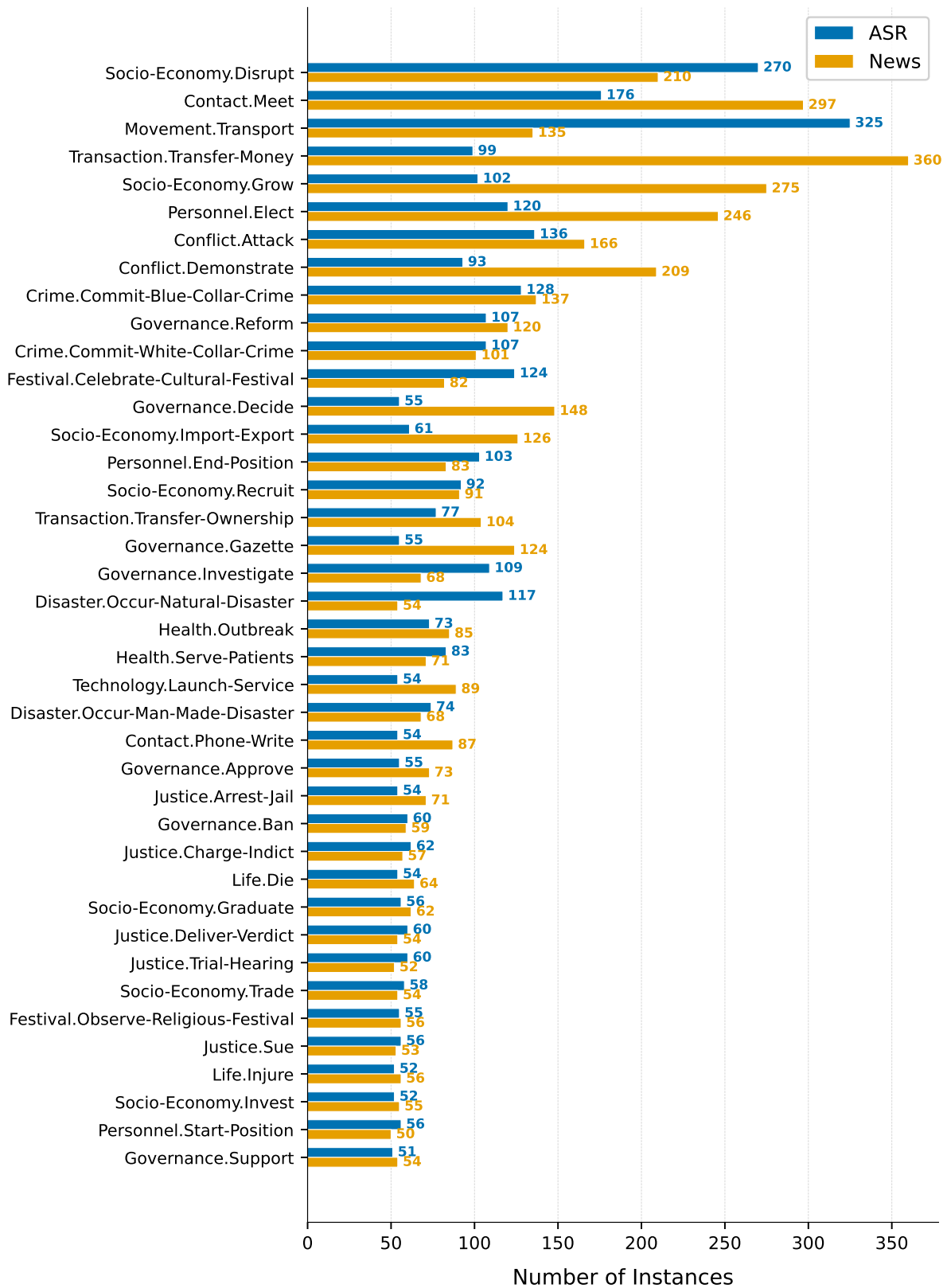


Figure 8: Distribution of event instances across the 40 event subtypes in the News and ASR corpora. Both corpora exhibit similar long-tail distributions, reflecting our topic-balancing effort despite the ASR corpus containing naturally occurring transcription artifacts and out-of-distribution vocabulary.

## B Prompt Design and Model Training

### B.1 Instruction Tuning Prompt Formats

The following example illustrates the structured code generation prompt under the **w/ Guide** setting, where fine-grained annotation guidelines (event definitions, trigger examples, and boundary constraints) are embedded directly within the schema representation as Python docstrings. English translations are provided in brackets for reference.

#### Instruction Tuning Prompt (w/ Guide)

```
# Instruction
Extract events according to the
schema.

# Schema
@dataclass
class Injure(Life):
    """
    এই ইভেন্টটি তখন ঘটে যখন
    অনিচ্ছাকৃত/স্বেচ্ছায়/দুর্ঘটনাবশত কোন
    ব্যক্তি শারীরিকভাবে আঘাত পান।
    (This event occurs when a person
    is physically injured unintention-
    ally/voluntarily/accidentally.)

    উদাহরণ ট্রিগার: আঘাত, ক্ষতি, আহত,
    অঙ্গহানি ইত্যাদি।
    (Example triggers: injure, harm,
    injured, dismemberment, etc.)

    সতর্কতা: (Note:)
    - আঘাতপ্রাপ্ত ব্যক্তি মারা গেলে তা এই
    ইভেন্টের অন্তর্ভুক্ত হবে না।
    (If the injured person dies, it will
    not be included in this event.)
    """
    mention: str

# Sentence
text = "আহত শিক্ষার্থীকে উদ্ধার করা সম্ভব
হয়েছে।"
(The injured student was
successfully rescued.)

result =

# Output
[
    Injure(mention="আহত")
    ("injured")
]
```

#### Instruction Tuning Prompt (w/o Guide)

```
# Instruction
Extract events according to the
schema.

# Schema
@dataclass
class Attack(Conflict):
    mention: str

# Sentence
text = "খুলনা বিশ্ববিদ্যালয়ে শিক্ষার্থী
নির্যাতনের অভিযোগে শিক্ষার্থীর সীট বাতিল
করেছে হল কর্তৃপক্ষ"
(The hall authority has cancelled
the seat of a student on
allegations of student
torture at Khulna
University.)

result =

# Output
[
    Attack(mention="নির্যাতনের")
    ("torture")
]
```

### B.2 Training Details

**Encoders** Encoder-only models are fine-tuned using full-parameter supervised fine-tuning with the AdamW optimizer (Loshchilov and Hutter, 2019), a learning rate of  $2 \times 10^{-5}$ , weight decay of 0.01, warmup over 10% of training steps, and a maximum sequence length of 128 tokens. Training is conducted for up to 15 epochs with early stopping based on validation Macro-F1 with patience 3.

**Decoders** Decoder-only LLMs are adapted via Low-Rank Adaptation (Hu et al., 2021) with rank  $r = 16$ , scaling factor  $\alpha = 32$ , and dropout  $p = 0$ , applied to all linear projection layers. We use a uniform rank across all model sizes following Al Monsur et al. (2026), who demonstrate that ranks above 8 provide no additional benefit for event detection tasks. Models are trained for up to 5 epochs with a learning rate of  $1 \times 10^{-4}$ , cosine learning rate schedule, warmup ratio of 0.05, effective batch size of 64 via gradient accumulation, and early stopping on validation loss with patience 3. All experiments are conducted on a single NVIDIA A100 GPU.

## C Comprehensive Event Ontology Guidelines

Table 5: Detailed definitions and examples for all 40 event subtypes.

| Event Type | Sub Type    | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Example                                                                                                                                                                                                          |
|------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contact    | Meet        | যখন দুই বা ততোধিক ব্যক্তি, দল বা প্রতিষ্ঠানের প্রতিনিধিরা সামনাসামনি সাক্ষাৎ, বৈঠক, সভা বা সম্মেলনে অংশগ্রহণ করেন।<br>(When two or more individuals, groups, or organizational representatives participate in a face-to-face meeting, conference, gathering, or discussion.)                                                                                                                                                                                                                                                                                            | সমসাময়িক বিষয় নিয়ে জরুরি এ সংবাদ <b>সম্মেলন</b> ডাকা হয়।<br>(An emergency press <b>conference</b> was called regarding the current situation.)                                                               |
|            | Phone-Write | যখন দুই বা ততোধিক ব্যক্তি, দল বা প্রতিষ্ঠানের প্রতিনিধিরা সামনাসামনি সাক্ষাতের পরিবর্তে ফোন, ইমেইল, চিঠি বা অন্যান্য লিখিত বার্তার মাধ্যমে যোগাযোগ বা আলোচনা করেন।<br>(When two or more individuals, groups, or organizational representatives communicate or discuss through phone calls, emails, letters, or other written messages instead of meeting face-to-face.)                                                                                                                                                                                                 | বিএনপির ভারপ্রাপ্ত চেয়ারম্যান চার নেতাকে <b>ফোন</b> করে নির্বাচনী প্রস্তুতি সম্পর্কে জানতে চান।<br>(The Acting Chairman of the BNP <b>called</b> four party leaders to inquire about the election preparations) |
| Conflict   | Attack      | যখন কোনো ব্যক্তি বা গোষ্ঠীর বিরুদ্ধে শারীরিক আক্রমণ, হামলা, সংঘর্ষ, যুদ্ধ, অভ্যুত্থান বা গুলিবর্ষণের মতো সহিংস ঘটনা ঘটে। আঘাতের নির্দিষ্ট মাধ্যম উল্লেখ না থাকলেও, যদি ঘটনাটি শারীরিক সহিংসতা বা সন্ত্রাসী কর্মকাণ্ড নির্দেশ করে, তবে সেটি <b>Attack</b> হিসেবে বিবেচিত হবে।<br>(When an individual or group is involved in a physically violent incident such as an attack, assault, clash, armed conflict, coup, or shooting. The specific means of injury need not be explicitly stated, provided the event describes physical violence or terrorist-like activity.) | ব্রাহ্মণবাড়িয়ায় আধিপত্য দ্বন্দ্ব দুই পক্ষের রক্তক্ষয়ী <b>সংঘর্ষ</b> হয়।<br>(A deadly <b>clash</b> occurred between two rival groups over dominance in Brahmanbaria.)                                        |
|            | Demonstrate | যখন কোনো ব্যক্তি বা জনগোষ্ঠী কোনো দাবি, অধিকার বা নির্দিষ্ট উদ্দেশ্য বাস্তবায়নের লক্ষ্যে সংগঠিতভাবে বিক্ষোভ, আন্দোলন, মানববন্ধন, ধর্মঘট, হরতাল বা অবরোধে অংশগ্রহণ করে।<br>(When an individual or group collectively participates in a protest, demonstration, human chain, strike, hartal, or blockade to express demands, grievances, or support for a particular cause.)                                                                                                                                                                                             | জনগণ ২৯ তারিখের <b>হরতাল</b> স্বপ্রণোদিত হয়ে পালন করেছে।<br>(The public voluntarily observed the <b>hartal</b> on the 29th.)                                                                                    |

| Event Type | Sub Type                  | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Example                                                                                                                                                                              |
|------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Crime      | Commit-Blue-Collar-Crime  | যখন নিম্ন আয়ের ব্যক্তি বা গোষ্ঠী কর্তৃক সংঘটিত প্রচলিত অপরাধ, যেমন হত্যা, ডাকাতি, ছিনতাই, অপহরণ, ধর্ষণ, চোরাচালান, মানবপাচার বা অগ্নিসংযোগের ঘটনা স্পষ্টভাবে উল্লেখ থাকে। অপরাধের ধরন অবশ্যই সুস্পষ্টভাবে উল্লেখ থাকতে হবে; শুধুমাত্র সংঘর্ষ, দ্বন্দ্ব, মারামারি বা গুলিবর্ষণের কথা বলা থাকলে সেটি <i>Conflict.Attack</i> হবে।<br>(When a conventional crime committed by an individual or group of lower socioeconomic background, such as murder, robbery, mugging, kidnapping, rape, smuggling, human trafficking, or arson, is explicitly mentioned. The type of crime must be clearly specified; if the text only describes a clash, conflict, fight, or shooting without explicitly identifying a crime, it should be Conflict.Attack.) | রাজধানীতে <b>ছিনতাই</b> রোধে শেষ রাতে টহল বাড়ানোর নির্দেশনা দেওয়া হয়েছে।<br>(Authorities have instructed increased late-night patrols in the capital to prevent <b>mugging</b> .) |
|            | Commit-White-Collar-Crime | যখন কোনো ব্যক্তি বা গোষ্ঠী দুর্নীতি, ঘুষ, প্রতারণা, জালিয়াতি, আত্মসাৎ, সাইবার অপরাধ, মানি লন্ডারিং বা অর্থ পাচারের মতো আর্থিক বা পেশাগত অপরাধে জড়িত থাকে। তবে প্রতারণা বা জালিয়াতির মতো অপরাধ সমাজের নিম্ন আয়ের ব্যক্তি কর্তৃক সংঘটিত হলেও <i>Commit-White-Collar-Crime</i> হিসেবে বিবেচিত হবে।<br>(When an individual or group commits a financial or occupational crime such as corruption, bribery, fraud, forgery, embezzlement, cybercrime, money laundering, or illicit financial transfer. Fraud and forgery committed by individuals from any socioeconomic background should also be annotated as Commit-White-Collar-Crime.)                                                                                                     | তার বিরুদ্ধে চেক <b>জালিয়াতির</b> অভিযোগে একাধিক মামলা হয়েছে।<br>(Multiple cases have been filed against him on charges of check <b>forgery</b> .)                                 |
| Disaster   | Occur-Man-Made-Disaster   | যখন মানবসৃষ্ট বা অনিচ্ছাকৃত দুর্যোগ, যেমন অগ্নিকাণ্ড, বিস্ফোরণ, সড়ক দুর্ঘটনা, ভবন ধস বা অনুরূপ দুর্ঘটনা সংঘটিত হয়।<br>(When a human-induced or accidental disaster, such as a fire, explosion, road accident, building collapse, or similar incident, occurs.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | শুক্রবার দুপুর সোয়া দুইটায় ঢাকার হাজারীবাগে চামড়ার গুদামে <b>আগুন</b> লাগে।<br>(A fire broke out at a leather warehouse in Hazaribagh, Dhaka, at around 2:15 p.m. on Friday.)     |
|            | Occur-Natural-Disaster    | যখন বন্যা, ঘূর্ণিঝড়, ভূমিকম্প, ভূমিধস, জলোচ্ছ্বাস, নদীভাঙন বা অন্যান্য প্রাকৃতিক দুর্যোগ সংঘটিত হয় বা তার প্রভাবের কথা উল্লেখ থাকে।<br>(When a natural disaster, such as a flood, cyclone, earthquake, landslide, storm surge, river erosion, or similar natural hazard, occurs or its impact is described.)                                                                                                                                                                                                                                                                                                                                                                                                                                 | সুরমা নদীর পানিতে <b>প্লাবিত</b> হয়েছে নদীতীরবর্তী গ্রামগুলো।<br>(The riverside villages have been <b>flooded</b> by the water of the Surma River.)                                 |

| Event Type | Sub Type                    | Definition                                                                                                                                                                                                                                                                                                                                            | Example                                                                                                                                                                                                                                |
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| Festival   | Celebrate-Cultural-Festival | যখন কোনো সাংস্কৃতিক উৎসব, জাতীয় দিবস, মেলা বা ঐতিহ্যবাহী অনুষ্ঠান উদযাপন, আয়োজন বা পালন করা হয়।<br>(When a cultural festival, national day, fair, or traditional celebration is organized, observed, or celebrated.)                                                                                                                               | দুই দেশই কূটনৈতিক সম্পর্কের সুবর্ণজয়ন্তী <b>উদযাপন</b> করেছে।<br>(Both countries <b>celebrated</b> the golden jubilee of their diplomatic relations.)                                                                                 |
|            | Observe-Religious-Festival  | যখন কোনো ধর্মীয় উৎসব, ধর্মীয় অনুষ্ঠান বা ধর্মীয় আচার-অনুষ্ঠান, যেমন ঈদ, দুর্গাপূজা, বুদ্ধ পূর্ণিমা, বড়দিন বা অনুরূপ ধর্মীয় দিবস পালন বা উদযাপন করা হয়।<br>(When a religious festival, ceremony, or observance, such as Eid, Durga Puja, Buddha Purnima, Christmas, or similar religious occasions, is celebrated or observed.)                  | <b>বৌদ্ধ পূর্ণিমা</b> উপলক্ষে আজ শান্তি শোভাযাত্রা অনুষ্ঠিত হয়েছে।<br>(A peace procession was held today to mark <b>Buddha Purnima</b> .)                                                                                             |
| Governance | Approve                     | যখন কোনো সরকার, সরকারি সংস্থা বা প্রতিষ্ঠান কোনো আইন, প্রকল্প, ঋণ, বাজেট, বরাদ্দ বা নীতিগত সিদ্ধান্ত আনুষ্ঠানিকভাবে অনুমোদন করে।<br>(When a government, public agency, or organization formally approves a law, project, loan, budget, allocation, or policy.)                                                                                        | মাধ্যমিক শিক্ষায় ৩০ কোটি ডলার ঋণ <b>অনুমোদন</b> করেছে বিশ্বব্যাংক।<br>(The World Bank <b>approved</b> a \$300 million loan for secondary education.)                                                                                  |
|            | Ban                         | যখন কোনো সরকার, সরকারি সংস্থা বা প্রতিষ্ঠান আদালতের আদেশ ব্যতীত কোনো ব্যক্তি, প্রতিষ্ঠান, কার্যক্রম, পণ্য বা সেবার ওপর নিষেধাজ্ঞা আরোপ বা কালোতালিকাভুক্ত করে।<br>(When a government, public agency, or organization imposes a ban or blacklists a person, organization, activity, product, or service, excluding bans issued through a court order.) | বাংলাদেশের ব্যাংকগুলোর সব ধরনের যানবাহন কেনায় <b>নিষেধাজ্ঞা</b> দিয়েছে কেন্দ্রীয় ব্যাংক।<br>(The central bank has imposed a <b>ban</b> on all vehicle purchases by banks in Bangladesh.)                                            |
|            | Decide                      | যখন কোনো সরকার, সরকারি সংস্থা, প্রতিষ্ঠান, পরিচালনা পর্ষদ বা অনুমোদিত কর্তৃপক্ষ আনুষ্ঠানিকভাবে কোনো সিদ্ধান্ত গ্রহণ করে।<br>(When a government, public agency, organization, board of directors, or other authorized authority formally makes a decision.)                                                                                            | বাজুসের কার্যনির্বাহী কমিটির সদস্যদের সর্বসম্মত মতামতের ভিত্তিতে দাম বাড়ানোর <b>সিদ্ধান্ত</b> হয়েছে।<br>(Based on the unanimous opinion of the BAJUS executive committee members, a <b>decision</b> was made to increase the price.) |
|            | Gazette                     | যখন কোনো সরকার, সরকারি সংস্থা বা প্রতিষ্ঠান আনুষ্ঠানিকভাবে প্রজ্ঞাপন, গেজেট বা নোটিশ জারি করে।<br>(When a government, public agency, or organization officially issues a gazette notification, proclamation, or formal notice.)                                                                                                                       | সোমবার এ-সংক্রান্ত <b>প্রজ্ঞাপন</b> জারি করেছে শিক্ষা মন্ত্রণালয়।<br>(The Ministry of Education issued a <b>gazette notification</b> regarding the matter on Monday.)                                                                 |

| Event Type | Sub Type       | Definition                                                                                                                                                                                                                                                                                                                                                                               | Example                                                                                                                                                                                                     |
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|            | Investigate    | যখন কোনো অভিযোগ, ঘটনা বা অনিয়মের সত্যতা যাচাইয়ের উদ্দেশ্যে সরকার, তদন্তকারী সংস্থা বা কোনো প্রতিষ্ঠান আনুষ্ঠানিকভাবে তদন্ত বা অনুসন্ধান পরিচালনা করে।<br>(When a government, investigative authority, or organization formally conducts an investigation or inquiry to verify the facts surrounding an allegation, incident, or irregularity.)                                         | দুর্নীতির অভিযোগে দুদক ঘটনাটি তদন্ত করছে।<br>(The Anti-Corruption Commission is <b>investigating</b> the incident following allegations of corruption.)                                                     |
|            | Reform         | যখন কোনো সরকার, সরকারি সংস্থা বা প্রতিষ্ঠান জনকল্যাণের উদ্দেশ্যে কোনো নীতি, আইন, বিধিমালা বা প্রশাসনিক ব্যবস্থার প্রণয়ন, পরিবর্তন বা সংস্কার করে।<br>(When a government, public agency, or organization formulates, revises, or reforms a policy, law, regulation, or administrative system intended to improve governance or public welfare.)                                          | নিরাপদ খাদ্য কর্তৃপক্ষ ট্রান্সফ্যাট নিয়ন্ত্রণ নীতি প্রণয়নের উদ্যোগ নিয়েছে।<br>(The Bangladesh Food Safety Authority has initiated the <b>formulation</b> of a policy to regulate trans fats.)            |
|            | Support        | যখন কোনো সরকার, সরকারি সংস্থা বা প্রতিষ্ঠান কোনো ব্যক্তি, গোষ্ঠী বা প্রতিষ্ঠানকে নগদ অর্থ, ভতুর্কি, ঋণ, ত্রাণ বা অর্থমূল্যসম্পন্ন অন্যান্য সহায়তা প্রদান করে।<br>(When a government, public agency, or organization provides financial or material assistance, such as subsidies, loans, cash, relief, or other support with monetary value, to an individual, group, or organization.) | ১০ টাকা কেজি চাল কর্মসূচিতে সরকার ২ হাজার ২০০ কোটি টাকা ভতুর্কি দিচ্ছে।<br>(The government is providing <b>subsidies</b> worth BDT 22 billion for the rice program that sells rice at BDT 10 per kilogram.) |
| Health     | Outbreak       | যখন কোনো সংক্রামক বা অসংক্রামক রোগের প্রাদুর্ভাব ঘটে, কোনো ব্যক্তি বা জনগোষ্ঠী আক্রান্ত হয়, রোগ শনাক্ত হয়, অথবা রোগের সংক্রমণের ঘটনা উল্লেখ থাকে।<br>(When a disease outbreak occurs, an individual or population is infected, a disease is diagnosed, or the transmission of a disease is reported.)                                                                                  | সম্প্রতি দেশে ডেঙ্গু রোগের প্রাদুর্ভাব বৃদ্ধি পেয়েছে।<br>(Recently, the <b>outbreak</b> of dengue has increased across the country.)                                                                       |
|            | Serve-Patients | যখন কোনো চিকিৎসক, হাসপাতাল বা স্বাস্থ্যসেবা প্রতিষ্ঠান রোগীকে চিকিৎসা, ভর্তি, পরামর্শ বা অন্যান্য স্বাস্থ্যসেবা প্রদান করে।<br>(When a doctor, hospital, or healthcare institution provides medical treatment, hospital admission, consultation, or other healthcare services to a patient.)                                                                                             | বর্তমানে তিনি আইসিইউতে চিকিৎসাধীন রয়েছেন।<br>(He is currently <b>undergoing treatment</b> in the intensive care unit (ICU)).                                                                               |

| Event Type | Sub Type        | Definition                                                                                                                                                                                                                                                                                                                                                                                        | Example                                                                                                                                                                                                                                                                                                                 |
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| Justice    | Arrest-Jail     | যখন পুলিশ, আইনশৃঙ্খলা রক্ষাকারী বাহিনী বা অন্য কোনো রাষ্ট্রীয় কর্তৃপক্ষ আইনগতভাবে কোনো অভিযুক্ত ব্যক্তিকে গ্রেফতার বা আটক করে, অথবা আদালতের আদেশে কারাগারে প্রেরণ বা কারাদণ্ড কার্যকর করা হয়।<br>(When a police force, law enforcement agency, or other government authority legally arrests or detains an accused person, or when an individual is imprisoned or sent to jail by court order.) | এ ঘটনায় বিএনপির আট জনকে <b>গ্রেফতার</b> করা হয়।<br>(Eight BNP members were <b>arrested</b> in connection with the incident.)                                                                                                                                                                                          |
|            | Charge-Indict   | যখন কোনো ব্যক্তি বা প্রতিষ্ঠানের বিরুদ্ধে আনুষ্ঠানিকভাবে কোনো অপরাধ বা অনিয়মের অভিযোগ আনা হয়, অথবা তাকে অভিযুক্ত করা হয়।<br>(When an individual or organization is formally charged, accused, or indicted for a crime or wrongdoing. The allegation does not need to be proven.)                                                                                                               | প্রাথমিকভাবে <b>অভিযুক্ত</b> পাঁচজনকে সাময়িক বহিষ্কার করা হয়েছে।<br>(The five individuals initially <b>accused</b> were temporarily suspended.)                                                                                                                                                                       |
|            | Deliver-Verdict | যখন কোনো আদালত কোনো মামলার রায়, আদেশ বা নির্দেশ প্রদান করে।<br>(When a court delivers a verdict, judgment, or judicial order in a legal case.)                                                                                                                                                                                                                                                   | মনিপুর স্কুলের বিষয়ে আদালতের <b>রায়</b> মেনে নেওয়ার আশ্বাস দিয়েছেন প্রতিমন্ত্রী।<br>(The State Minister assured that the court's <b>verdict</b> regarding Monipur School would be accepted.)                                                                                                                        |
|            | Sue             | যখন কোনো ব্যক্তি, প্রতিষ্ঠান বা সংস্থা আদালতে অন্য ব্যক্তি বা প্রতিষ্ঠানের বিরুদ্ধে মামলা দায়ের করে।<br>(When an individual, organization, or institution files a lawsuit or legal case against another individual or organization.)                                                                                                                                                             | ভুক্তভোগী অভিযুক্তদের বিরুদ্ধে <b>মামলা</b> করেছেন।<br>(The victim has filed a <b>lawsuit</b> against the accused.)                                                                                                                                                                                                     |
|            | Trial-Hearing   | যখন কোনো আদালত কোনো মামলার আনুষ্ঠানিক শুনানি, জামিন শুনানি বা বিচারিক কার্যক্রম পরিচালনা করে।<br>(When a court conducts a formal trial, hearing, bail hearing, or other judicial proceeding related to a legal case.)                                                                                                                                                                             | বিশ্ববিদ্যালয়ের ক্যাম্পাসের জমি কেনা-সংক্রান্ত অর্থ আত্মসাতের মামলায় ট্রাস্টি বোর্ডের চার সদস্যের জামিন <b>শুনানি</b> রোববার অনুষ্ঠিত হবে।<br>(The bail <b>hearing</b> for four members of the Board of Trustees in the embezzlement case related to the purchase of land for the University will be held on Sunday.) |

| Event Type | Sub Type       | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Example                                                                                                                                                                                                      |
|------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Life       | Die            | যখন কোনো ব্যক্তি স্বাভাবিক কারণে, দুর্ঘটনায়, সহিংস ঘটনায় বা আত্মত্যাগের ফলে মৃত্যুবরণ করেন।<br>মৃত্যুর ফলাফল নির্দেশকারী শব্দ, যেমন নিহত, মৃত, মারা যান বা প্রাণ হারান উল্লেখ থাকলেও এটি <i>Die</i> ইভেন্ট হিসেবে বিবেচিত হবে।<br>(When a person dies due to natural causes, an accident, violence, or self-sacrifice. Resultative expressions indicating death, such as killed, dead, died, or lost one's life, should also be annotated as Die events.) | নিহত শাহীনের পরিবারের পাশে দাঁড়াতে নেতাকর্মীদের প্রতি আহ্বান জানানো হয়।<br>(Party leaders and activists were urged to stand beside the family of the <b>deceased</b> Shahin.)                              |
|            | Injure         | যখন কোনো ব্যক্তি দুর্ঘটনা, সহিংসতা বা অন্য কোনো কারণে শারীরিকভাবে আহত বা আঘাতপ্রাপ্ত হন।<br>(When a person sustains a physical injury due to an accident, violence, or any other cause. If the person dies as a result of the incident, the event should be annotated as Die, not Injure.)                                                                                                                                                                  | গাছের সঙ্গে ধাক্কা লেগে বিশ্ববিদ্যালয়ের অন্তত ১৬ শিক্ষার্থী আহত হন।<br>(At least 16 university students were <b>injured</b> after their vehicle collided with a tree.)                                      |
| Movement   | Transport      | যখন কোনো ব্যক্তি, যানবাহন বা পণ্য এক স্থান থেকে অন্য স্থানে পরিবহন, যাত্রা, স্থানান্তর বা রওনা হয়।<br>(When a person, vehicle, or goods are transported, travel, depart, or move from one location to another.)                                                                                                                                                                                                                                            | তিনি বিমান বাংলাদেশের একটি ফ্লাইটে চিকিৎসার জন্য ভারতের চেন্নাইয়ের উদ্দেশ্যে <b>রওনা</b> দিয়েছেন।<br>(He <b>departed</b> for Chennai, India, on a Biman Bangladesh Airlines flight for medical treatment.) |
| Personnel  | Elect          | যখন কোনো ব্যক্তি নির্বাচনের মাধ্যমে কোনো পদে নির্বাচিত হন, অথবা কোনো পদে প্রতিনিধি নির্বাচনের উদ্দেশ্যে নির্বাচন অনুষ্ঠিত বা আয়োজন করা হয়।<br>(When a person is elected to a position through an election, or when an election is held or organized to select representatives for a position.)                                                                                                                                                            | ঢাকা চেম্বার অব কমার্স অ্যান্ড ইন্ডাস্ট্রির (ডিসিসিআই) সভাপতি <b>নির্বাচিত</b> হয়েছেন তিনি।<br>(He was <b>elected</b> President of the Dhaka Chamber of Commerce and Industry.)                             |
|            | End-Position   | যখন কোনো ব্যক্তি চাকরি, পদ বা দায়িত্ব থেকে বরখাস্ত, পদচ্যুত, ছাঁটাই, অব্যাহতি, পদত্যাগ বা অবসরের মাধ্যমে দায়িত্ব থেকে সরে যান।<br>(When a person leaves a job or position through dismissal, termination, removal, resignation, retirement, or the end of their tenure.)                                                                                                                                                                                  | এক নারী কর্মীকে চাকরি থেকে <b>বরখাস্ত</b> করেছে চীনা ই-কমার্স প্রতিষ্ঠান আলিবাবা।<br>(The Chinese e-commerce company Alibaba <b>dismissed</b> a female employee from her job.)                               |
|            | Start-Position | যখন কোনো ব্যক্তি কোনো পদে নিয়োগপ্রাপ্ত হন, নতুন চাকরিতে যোগদান করেন বা আনুষ্ঠানিকভাবে দায়িত্ব গ্রহণ করেন।<br>(When a person is appointed to a position, joins a new job, or officially assumes a role or responsibility.)                                                                                                                                                                                                                                 | একদিনের চাকরি মেলায় তিনজন প্রার্থী <b>নিয়োগ</b> পেয়েছেন।<br>(Three candidates were <b>appointed</b> at the one-day job fair.)                                                                             |

| Event Type    | Sub Type      | Definition                                                                                                                                                                                                                                                                                                                                                     | Example                                                                                                                                                                                   |
|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Socio-Economy | Disrupt       | যখন অর্থনীতি, বাণিজ্য, শিল্প, উৎপাদন, সরবরাহ ব্যবস্থা, শিক্ষা, কর্মসংস্থান বা জনসেবাসংশ্লিষ্ট কোনো কার্যক্রম উল্লেখযোগ্যভাবে ব্যাহত, বিঘ্নিত বা স্থবির হয়।<br>(When activities related to the economy, trade, industry, production, supply chains, education, employment, or public services are significantly disrupted, interrupted, or brought to a halt.) | আর্থিক সংকট বিশ্বব্যাপী সরবরাহ চেইনকে <b>ব্যাহত</b> করেছে।<br>(The financial crisis has <b>disrupted</b> global supply chains.)                                                           |
|               | Graduate      | যখন কোনো ব্যক্তি বা আবেদনকারী কোনো শিক্ষাগত বা প্রতিযোগিতামূলক পরীক্ষায় উত্তীর্ণ হন, স্নাতক সম্পন্ন করেন বা আনুষ্ঠানিকভাবে শিক্ষাজীবনের একটি ধাপ সম্পন্ন করেন।<br>(When a person or applicant passes an academic or competitive examination, graduates from an educational program, or formally completes a stage of education.)                              | এতে চূড়ান্তভাবে <b>উত্তীর্ণ</b> হয়েছেন ২৩,৯৮৫ জন।<br>(A total of 23,985 candidates <b>passed</b> the examination.)                                                                      |
|               | Grow          | যখন অর্থনীতি, শিল্প, বাণিজ্য বা অন্যান্য পরিমাপযোগ্য সূচকের প্রবৃদ্ধি, বৃদ্ধি বা সম্প্রসারণের কথা উল্লেখ থাকে।<br>(When the growth, increase, or expansion of an economic indicator, industry, trade, or another measurable quantity is reported.)                                                                                                             | বাংলাদেশ গত ১৩ বছরে গড়ে ৬.৬% জিডিপি <b>প্রবৃদ্ধি</b> অর্জন করেছে।<br>(Bangladesh has achieved an average GDP <b>growth</b> of 6.6% over the past 13 years.)                              |
|               | Import-Export | যখন কোনো পণ্য বিদেশ থেকে দেশে আমদানি করা হয় অথবা দেশ থেকে বিদেশে রপ্তানি করা হয়।<br>(When goods are imported into a country from abroad or exported from a country to foreign markets.)                                                                                                                                                                      | ভোজ্যতেলের চাহিদার একটি অংশ <b>আমদানি</b> করা বীজ থেকে তেল উৎপাদনের মাধ্যমে পূরণ করা হয়।<br>(A portion of the demand for edible oil is met by producing oil from <b>imported</b> seeds.) |
|               | Invest        | যখন কোনো ব্যক্তি, প্রতিষ্ঠান বা রাষ্ট্র কোনো ব্যবসা, প্রকল্প, শিল্প বা দেশে অর্থ বা পুঁজি বিনিয়োগ করে।<br>(When an individual, organization, or government invests money or capital in a business, project, industry, or country.)                                                                                                                            | থাইল্যান্ডের বিনিয়োগকারীদের বাংলাদেশে <b>বিনিয়োগে</b> আকৃষ্ট করা জরুরি।<br>(It is important to attract Thai investors to <b>invest</b> in Bangladesh.)                                  |
|               | Recruit       | যখন কোনো প্রতিষ্ঠান বা সংস্থা নির্দিষ্ট পদের জন্য নিয়োগ বিজ্ঞপ্তি প্রকাশ করে বা আবেদন আহ্বান করে।<br>(When an organization or institution publishes a recruitment notice or invites applications for one or more positions.)                                                                                                                                  | জনবল <b>নিয়োগের বিজ্ঞপ্তি</b> দিয়েছে বাংলাদেশ সেনাবাহিনী।<br>(The Bangladesh Army has published a <b>recruitment notice</b> .)                                                          |

| Event Type  | Sub Type           | Definition                                                                                                                                                                                                                                                                                                                                                                                                 | Example                                                                                                                                                                                                         |
|-------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | Trade              | যখন সামগ্রিক বাণিজ্য কার্যক্রমের কথা সাধারণভাবে উল্লেখ থাকে।<br>(When the text generally refers to overall trade activities.)                                                                                                                                                                                                                                                                              | বড় ধরনের <b>বাণিজ্য</b> ঘাটতিতে পড়ে ক্রমাগত চলতি হিসাবের ঘাটতিও বেড়েছিল বাংলাদেশের।<br>(Bangladesh's current account deficit continued to increase as the country experienced a large <b>trade</b> deficit.) |
| Technology  | Launch-Service     | যখন কোনো বিদ্যমান প্রতিষ্ঠান বা সংস্থা নতুন কোনো প্রযুক্তিপণ্য, ডিজিটাল সেবা, অ্যাপ্লিকেশন বা অনুরূপ সেবা আনুষ্ঠানিকভাবে চালু বা বাজারে উন্মোচন করে।<br>(When an existing organization or institution officially launches or introduces a new technology product, digital service, application, or similar service.)                                                                                       | হজ অ্যাপ 'লাব্বাইক' চালু করল মন্ত্রণালয়।<br>(The ministry <b>launched</b> the 'Labbaik' Hajj application.)                                                                                                     |
| Transaction | Transfer-Money     | যখন কোনো ব্যক্তি, প্রতিষ্ঠান বা সংস্থা কোনো পণ্য বা সেবা ক্রয়ের উদ্দেশ্যে ব্যতীত অন্য ব্যক্তি বা প্রতিষ্ঠানের কাছে অর্থ প্রদান, গ্রহণ, ঋণ, অনুদান, দান বা আর্থিক সহায়তা প্রদান বা গ্রহণ করে।<br>(When an individual, organization, or institution transfers, receives, lends, borrows, donates, or grants money to another party without the primary purpose of purchasing goods or services.)           | বাণিজ্যিক ব্যাংকগুলো টাকা <b>ধার</b> করতে হলে আগের চেয়ে দশমিক ২৫ শতাংশ বেশি সুদ গুনতে হবে।<br>(Commercial banks will have to pay 0.25 percentage points more in interest to <b>borrow</b> money.)              |
|             | Transfer-Ownership | যখন কোনো ব্যক্তি, প্রতিষ্ঠান, যানবাহন, অস্ত্র, সম্পদ বা অন্যান্য মালামালের মালিকানা ক্রয়, বিক্রয়, হস্তান্তর, দান বা ধার দেওয়া-নেওয়ার মাধ্যমে এক পক্ষ থেকে অন্য পক্ষের কাছে স্থানান্তরিত হয়।<br>(When the ownership of a person-owned asset, organization, vehicle, weapon, property, or other goods is transferred from one party to another through purchase, sale, transfer, donation, or lending.) | সরকারের অনুমতি পেলে তারা আগে <b>ক্রয়কৃত</b> পেঁয়াজ বুধবার বাংলাদেশে পাঠাবেন।<br>(Subject to government approval, they will ship the <b>purchased</b> onions to Bangladesh on Wednesday.)                      |